

Four Advanced Solutions for Climate Classification

Extreme Class Imbalance Problem

Common Challenge

Base Model: DeBERTa-v3 | Goal: Prevent collapse, improve precision



Four Different Solution Approaches

SOLUTION 1

Data Augmentation + Focal Loss

- Multi-strategy augmentation
- Balanced dataset creation
 - Adaptive Focal Loss
- Threshold optimization
 - Cross-validation
 - Weight averaging

SOLUTION 2

Cost-Sensitive + SMOTE

- SMOTE augmentation
- Heavy class weighting
- Test-time augmentation
 - MC Dropout
 - Attention pooling
 - Calibration ready

SOLUTION 3

Multi-Model Ensemble

- DeBERTa + oversampling
- RoBERTa + undersampling
- DistilBERT + balanced
 - Weighted ensemble
 - Cross-validation
- Architectural diversity

SOLUTION 4

Contrastive Learning + Mixup

- Supervised contrastive loss
 - Token-level mixup
- Curriculum learning
- Multi-task learning
 - Projection head
 - Cross-validation

SOLUTION SELECTION

Choose based on: Computational resources | Publication goals | Performance needs



Key Advantages Over Baseline

- ✓ Substantially improved precision
 - ✓ Maintained high recall
- ✓ All themes predicted (no collapse)
 - ✓ Reduced over-acceptance
- ✓ Advanced imbalance handling
 - ✓ Better calibration